

Exam Date & Time: 14-Jun-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MIT BLR - BTech VI Semester Makeup Examination, June -2025

DATA WAREHOUSING AND MINING [CSE-4060]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) List the various Data Mining Functionalities with an example. (2)
 - A)
 - B) Two text documents are represented as term frequency vectors below. Compute the cosine similarity:
 $D1=(1,3,0,2,0,0,0)$ $D2=(0,0,0,2,1,0,3)$. (5)
 - C) Consider the following paired data:
 $X = [1, 2, 3]$
 $Y = [4, 5, 6]$
Compute the covariance between X and Y. (3)
- 2) A bank wants to evaluate customer satisfaction by collecting feedback. However, the customers are spread across five major cities, each with a different population size. Which sampling technique is most suitable in this case? Justify with proper implementation steps of the sampling process. (2)
 - A)
 - B) A retail company wants to move data from a MySQL database and web log files into a central data warehouse. Explain with a neat diagram how the ETL process helps in this scenario. Highlight one challenge from each phase: Extract, Transform, and Load. (5)
 - C) Discuss the OLAP operations and its characteristics used for data analysis in multidimensional databases? Illustrate each operation with a suitable diagram and example. (3)
- 3) Illustrate "Market Basket Analysis" technique with appropriate example. (5)
 - A)
 - B) (5)

TID	List
T100	I1,I2,I5
T100	I2,I4
T100	I2,I3
T100	I1,I2,I4
T100	I1,I3
T100	I2,I3
T100	I1,I3
T100	I1,I2,I3,I5
T100	I1,I2,I3

Consider the above database, consisting of 9 transactions. Suppose the absolute minimum support count required is 2 and with relative support $\text{min_sup} = 2/9 = 22\%$. Let minimum confidence required is 70% for association rules.

(a) Find out the frequent itemset using Apriori algorithm.

(b) Identify Association rules will be generated using min. support & min. confidence

4)

Construct a decision tree based on this training data. For splitting, use information gain as measure for impurity. Build a separate branch for each attribute.

A)

ID	Time	Gender	Area	Risk
1	1-2	M	Urban	Low
2	2-7	M	Rural	High
3	>7	F	Rural	Low
4	1-2	F	Rural	High
5	>7	M	Rural	High
6	1-2	M	Rural	High
7	2-7	F	Urban	Low
8	2-7	M	Urban	low

(5)

B)

(5)

Dataset:

Sample	X_1 (Weather)	X_2 (Temperature)	Y (Play Tennis)
S1	Sunny	Hot	No
S2	Sunny	Hot	No
S3	Overcast	Hot	Yes
S4	Rainy	Mild	Yes
S5	Rainy	Cool	Yes
S6	Rainy	Cool	No
S7	Overcast	Cool	Yes
S8	Sunny	Mild	No
S9	Sunny	Cool	Yes
S10	Rainy	Mild	Yes
S11	Sunny	Mild	Yes
S12	Overcast	Mild	Yes
S13	Overcast	Hot	Yes
S14	Rainy	Mild	No

You are given a new instance with $X_1 = \text{Sunny}$, $X_2 = \text{Cool}$

Using Naïve Bayes, calculate the probability that: $Y = \text{Yes}$. $Y = \text{No}$. Then classify the new instance.

- 5) Explain the working principles of two different Hierarchical methods of clustering.

(5)

A)

B)

- List two different Density based methods of clustering and describe its working principles.

(5)

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